

Counterfactual Analysis for Structural Dynamic Discrete Choice Models

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1 Goal

- Build validated replication code for a dynamic discrete-choice model with partially identified payoffs.
- Compare entry-cost and marginal-cost subsidies while carrying payoff uncertainty into every counterfactual outcome.

2 Setting and Data-Generating Process

2.1 Model setup

- A single firm chooses whether to be inactive, $a_t = 0$, or active, $a_t = 1$.
- The state is $x_t = (k_t, w_t)$, where $k_t = a_{t-1} \in \{0, 1\}$ and the observed profit state is $w_t \in \{L, H\}$.
- The profit state follows a known transition matrix P , independently of the current action, and $k_{t+1} = a_t$.
- The model has state-specific scrap values s_w , entry costs e_w , and running costs r_w . Entry and running costs are nonnegative.

Table 1: Deterministic flow payoffs

Previous action	Inactive, $a_t = 0$	Active, $a_t = 1$
$k_t = 0$	0	$\pi_w - r_w - e_w$
$k_t = 1$	s_w	$\pi_w - r_w$

The calibration uses $\pi = (2, 4)$, $\beta = 0.9$, a persistent two-state profit process, and true values $s = (4.5, 4.5)$, $e = (5, 5)$, and $r = (0.5, 0.5)$. Type-I extreme-value shocks generate logit conditional choice probabilities (CCPs).

2.2 Model solution

Let $V(k, w)$ denote the ex-ante value. The choice-specific values satisfy

$$v_a(k, w) = u(k, w, a) + \beta E[V(a, w') \mid w].$$

The extreme-value shocks imply the inclusive-value equation

$$V(k, w) = \log(\exp(v_0(k, w)) + \exp(v_1(k, w))),$$

and the active-choice probability is

$$p_{kw} = \frac{\exp(v_1(k, w))}{\exp(v_0(k, w)) + \exp(v_1(k, w))}.$$

Value iteration solves this four-state fixed point and returns the four population CCPs.

3 Identification

Let $p_{kw} = \Pr(a = 1 \mid k, w)$ and $d_{kw} = \log[p_{kw}/(1 - p_{kw})]$. For a fixed profit state, the continuation term is common across the two lagged-action states:

$$C_w = \beta \sum_{w'} P_{ww'} [V(1, w') - V(0, w')].$$

The two log-odds equations are

$$d_{0w} = \pi_w - r_w - e_w + C_w, \quad d_{1w} = \pi_w - r_w - s_w + C_w.$$

Taking their difference eliminates the common continuation value and gives the first identifying equation:

$$d_{1w} - d_{0w} = e_w - s_w.$$

For the second equation, compare the inactive paths $(0, w) \rightarrow (0, w')$ and $(1, w) \rightarrow (0, w')$. Their continuation values coincide. The logit inclusive-value identity therefore gives

$$V(1, w) - V(0, w) = s_w + \log\left(\frac{1 - p_{0w}}{1 - p_{1w}}\right) = s_w + A_w(p),$$

where $A_w(p) = \log[(1 - p_{0w})/(1 - p_{1w})]$. For each candidate $s = (s_L, s_H)$, the two equations recover

$$e_w(s; p) = s_w + d_{1w} - d_{0w},$$

$$r_w(s; p) = \pi_w - e_w(s; p) + \beta \sum_{w'} P_{ww'} [s_{w'} + A_{w'}(p)] - d_{0w}.$$

Nonnegative entry and running costs restrict the free scrap values to

$$\mathcal{S}(p) = \{s \in \mathbb{R}^2 : e_w(s; p) \geq 0, r_w(s; p) \geq 0, w \in \{L, H\}\}.$$

The payoff identified set is

$$\Theta_I(p) = \{(s, e(s; p), r(s; p)) : s \in \mathcal{S}(p)\}.$$

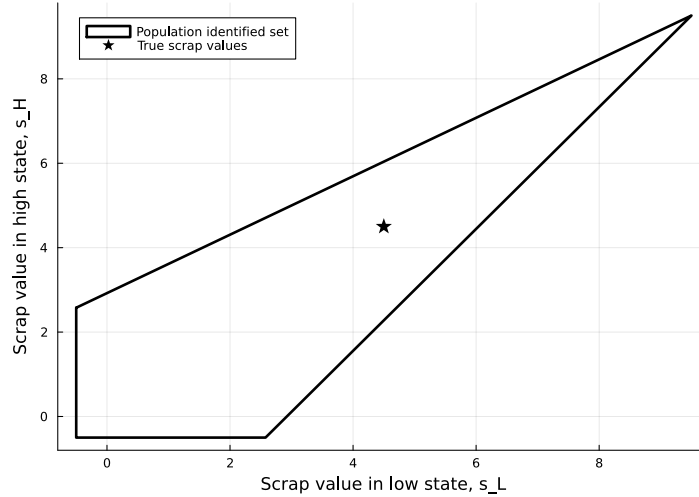


Figure 1: The population restrictions form a polygon in (s_L, s_H) . The star is the true payoff vector, which lies inside the set but is not point identified.

4 Estimation

4.1 Data-generating process

The simulation draws actions from the four population CCPs, sets $k_{t+1} = a_t$, and draws w_{t+1} from $P(\cdot | w_t)$. The sample contains 2,000 firms observed for 20 periods.

4.2 Summary Statistics

Table 2: State-level observations and active choices

k_t	Profit state	Observations	Active choices	Active share
0	Low	4232	3024	0.715
0	High	2504	2393	0.956
1	Low	15684	12562	0.801
1	High	17580	17048	0.97

4.3 Estimated CCPs

Each CCP is the active-choice frequency in state (k, w) :

$$\hat{p}_{kw} = \frac{\sum_{i,t} \mathbf{1}\{k_{it} = k, w_{it} = w, a_{it} = 1\}}{\sum_{i,t} \mathbf{1}\{k_{it} = k, w_{it} = w\}}.$$

Table 3: Estimated and population CCPs

k_t	Profit state	$\hat{p}_{kw}$	p_{kw}
0	Low	0.715	0.714
0	High	0.956	0.951
1	Low	0.801	0.805
1	High	0.97	0.97

4.4 Sample objective

Replacing p and P with $\hat{p}$ and $\hat{P}$ gives four sample restrictions $\hat{m}(\theta)$. With $W = I_4$, the minimum-distance objective is

$$\hat{Q}(\theta) = \hat{m}(\theta)' \hat{m}(\theta), \quad \theta = (s_L, s_H, e_L, e_H, r_L, r_H).$$

At the true payoff vector, the sample objective equals 0.04597; sampling error prevents it from being exactly zero.

4.5 Estimated identified set

Plugging $\hat{p}$ and $\hat{P}$ into the analytic formulas for $e(s)$ and $r(s)$ yields the estimated polygon $\hat{\mathcal{S}}$. A grid search checks its two-dimensional projection.

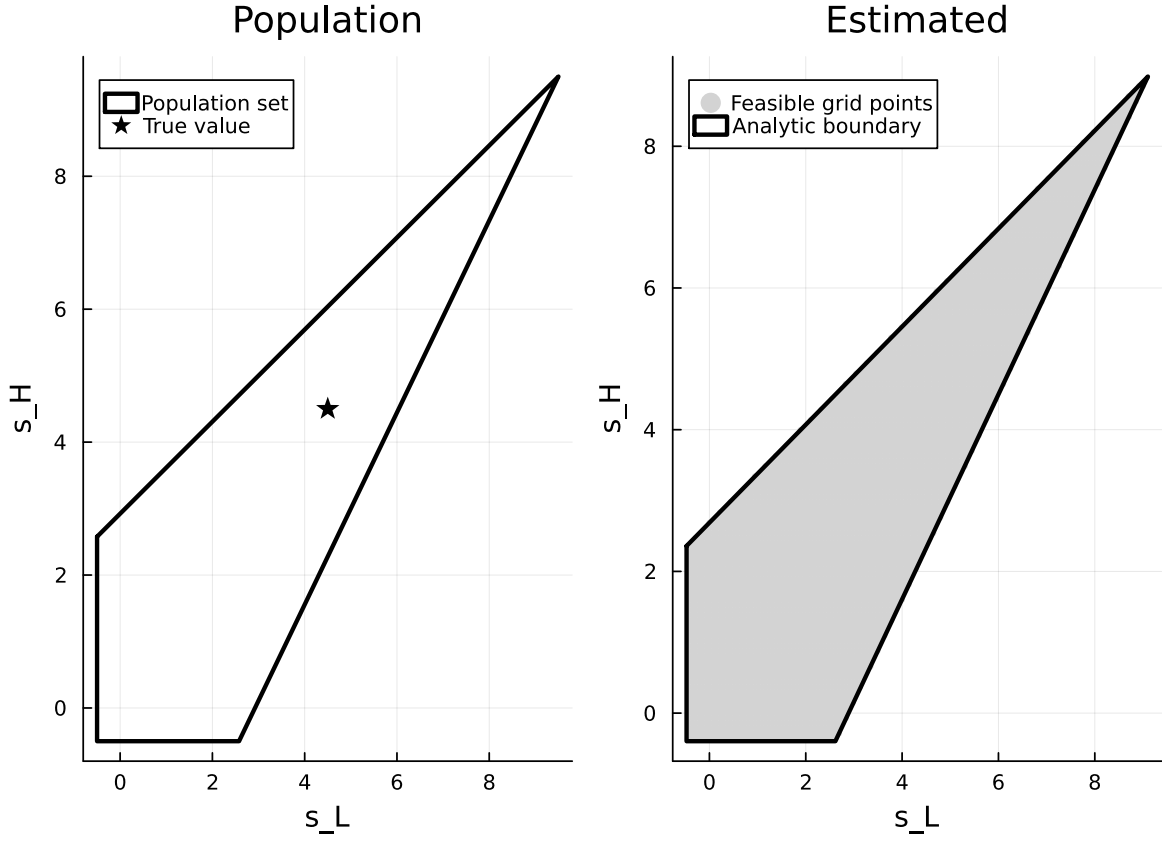


Figure 2: The population and estimated payoff sets. Gray grid points numerically reproduce the analytic estimated boundary.

5 Counterfactual Analysis

5.1 Overview

- The policy grid uses $\tau_E, \tau_M \in [0, 0.5]$. Entry subsidies are paid when an inactive firm enters; marginal-cost subsidies are paid whenever the firm operates.
- Every policy is solved for every admissible payoff vector in the estimated identified set. The minimum and maximum outcomes form numerical counterfactual bounds.
- The calculation covers the polygon interior, vertices, and edges. A denser grid checks the numerical approximation.

The proportional subsidies change entry and running costs to

$$e_w^{cf} = (1 - \tau_E)e_w, \quad r_w^{cf} = (1 - \tau_M)r_w.$$

Let μ_{kw}^{cf} denote the stationary state probability and p_{kw}^{cf} the active CCP. The long-run active share and its percentage change are

$$\mu_1^{cf} = \sum_{k,w} \mu_{kw}^{cf} p_{kw}^{cf}, \quad R_\mu = 100 \frac{\mu_1^{cf} - \mu_1^0}{\mu_1^0}.$$

Long-run expenditure per firm and period is

$$G_E = \sum_w \tau_E e_w \mu_{0w}^{cf} p_{0w}^{cf},$$

$$G_M = \sum_{k,w} \tau_M r_w \mu_{kw}^{cf} p_{kw}^{cf}, \quad G = G_E + G_M.$$

Cost-effectiveness is $CE = R_\mu/G$ when $G > 0$.

5.2 Entry Cost Subsidy

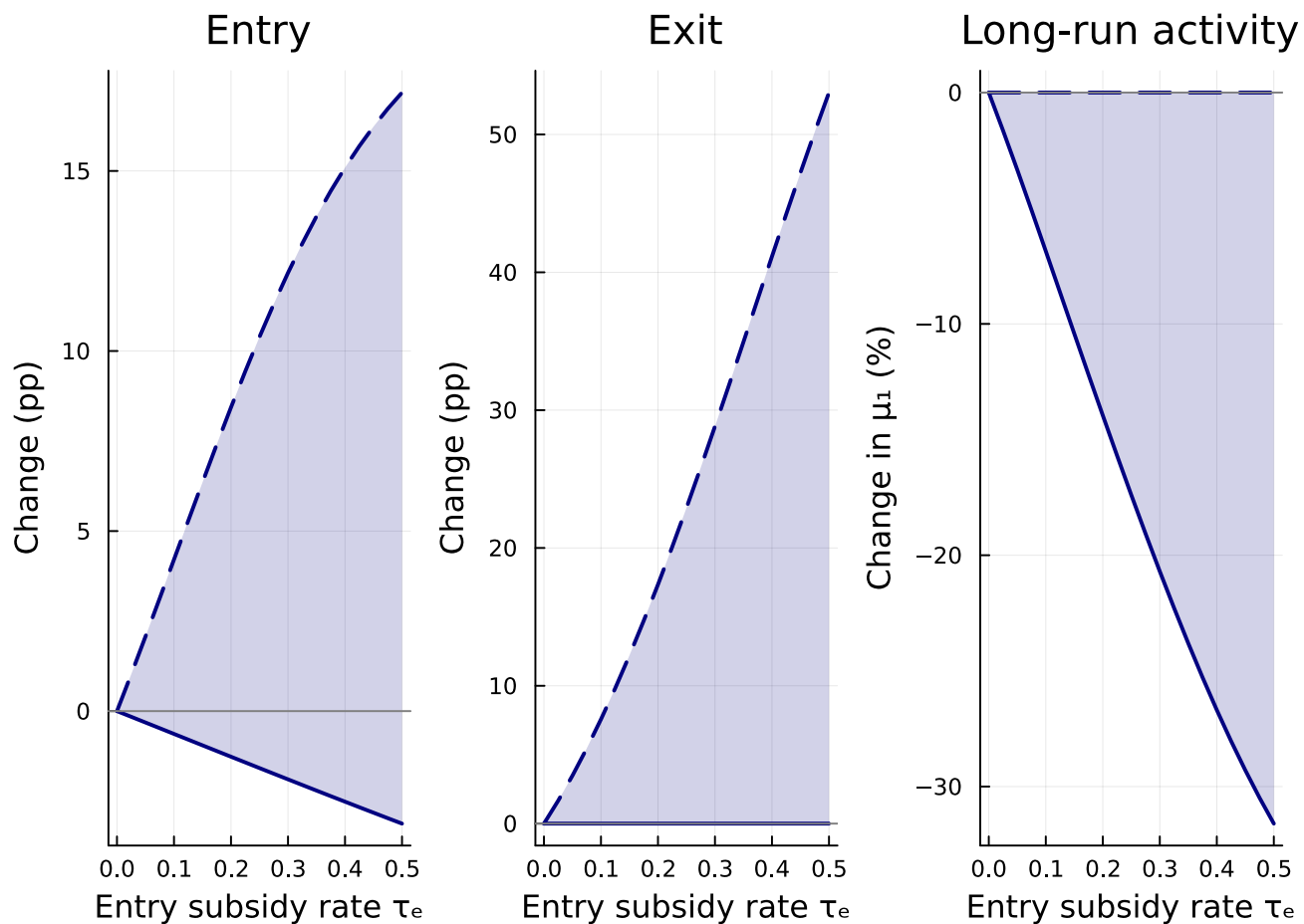


Figure 3: An entry-cost subsidy can increase exit as firms preserve the option to re-enter with support. Across the identified set, the exit response can dominate and reduce long-run activity.

5.3 Marginal Cost Subsidy

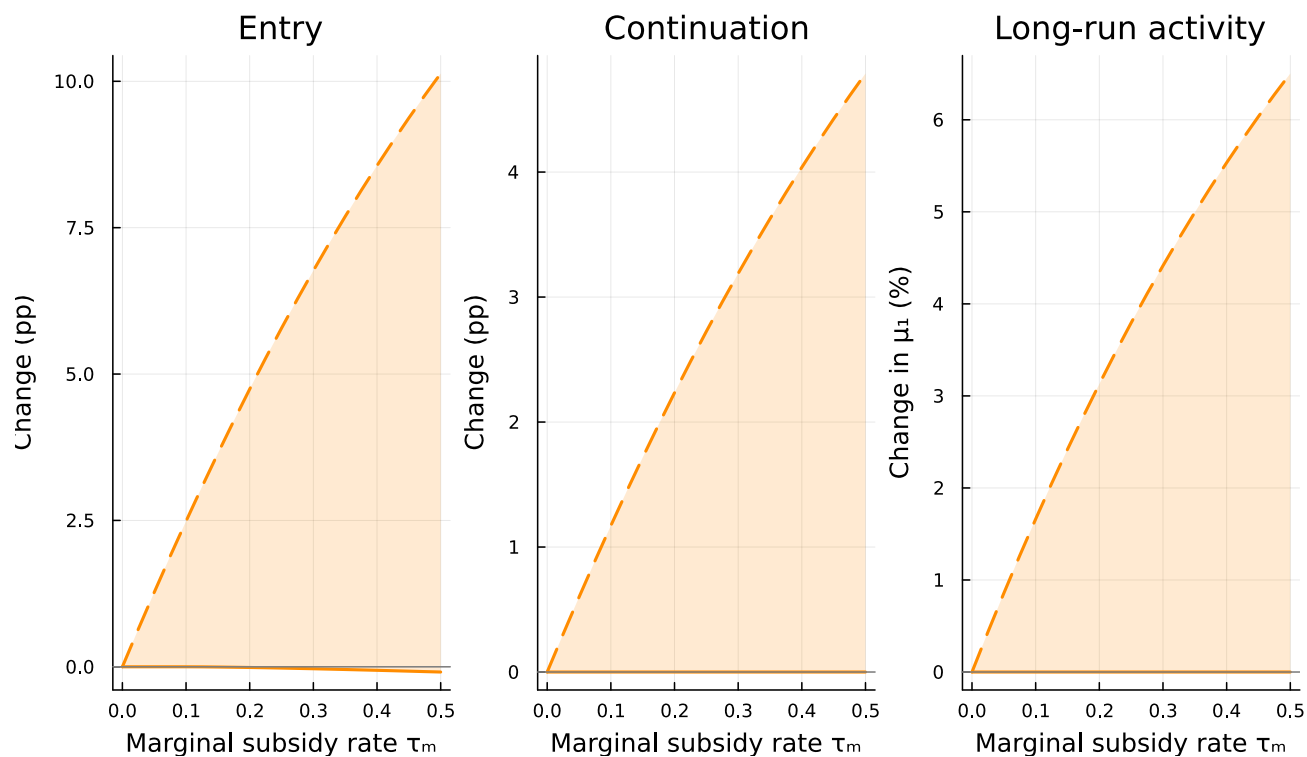


Figure 4: A marginal-cost subsidy supports both entry and continued operation. The long-run active-share response is weakly positive over the estimated payoff set.

5.4 Mixed Policy Design

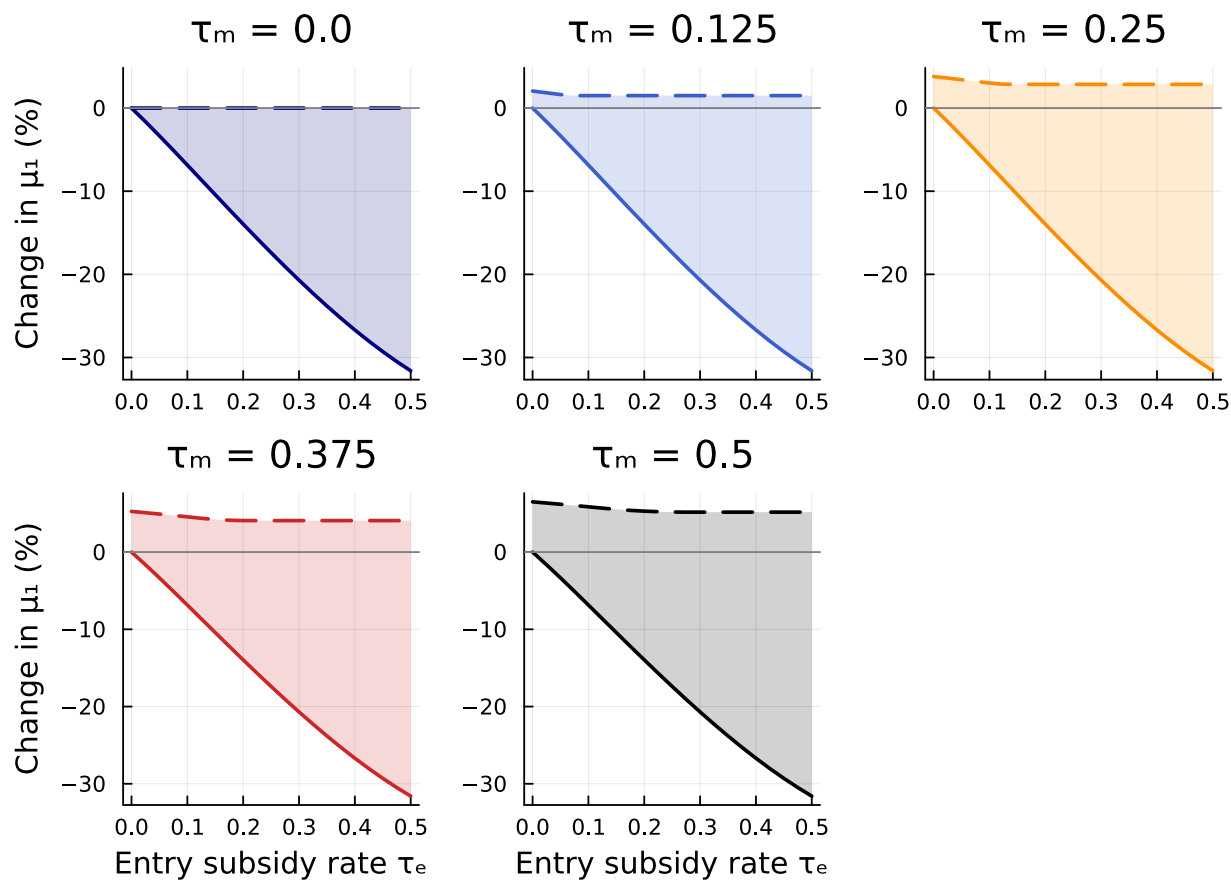


Figure 5: Identified bounds for the percentage change in the stationary active share across the two-policy grid.

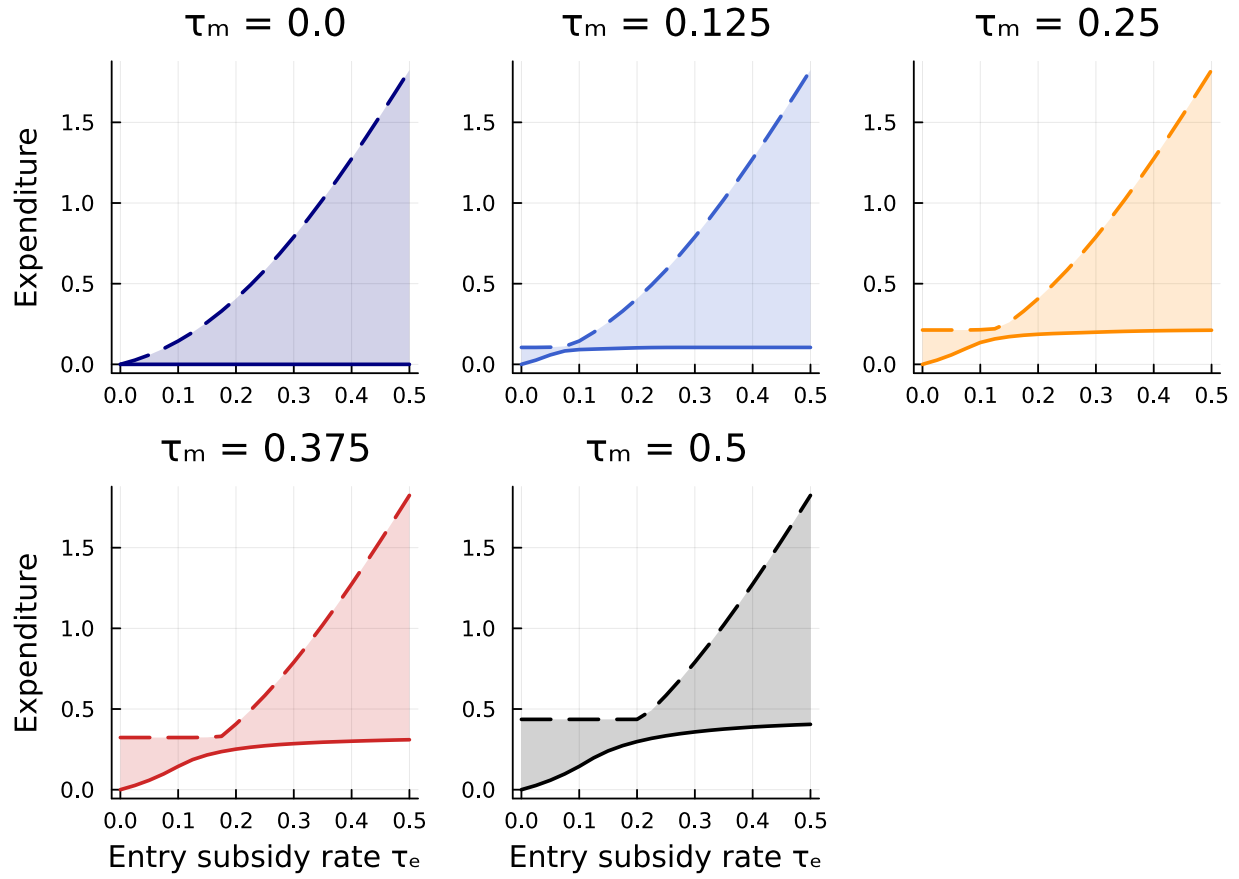


Figure 6: Identified expenditure bounds. Entry and marginal-cost subsidies differ in timing and payment frequency.

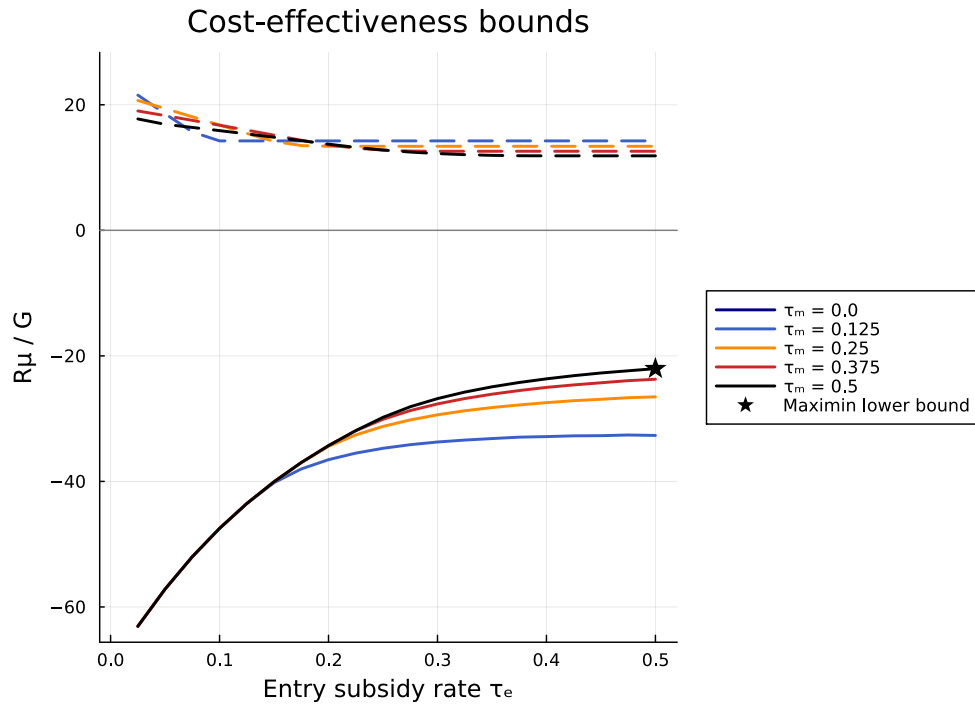


Figure 7: Lower and upper cost-effectiveness bounds. Each color fixes τ_M while τ_E varies; gaps fail the minimum-expenditure threshold.